

INCIDENT INVESTIGATION REPORT

Proxy Telemetry Analysis — HaveIBeenPwned / VirusTotal / zone-h.org Activity

Classification	Internal — Confidential
Case Reference	INC-2021-1116-01
Report Type	Digital Forensics & Incident Response (DFIR) — Log-Based Findings
Evidence Source	Blue Coat ProxySG Web Access Logs (sourcetype: bluecoat:proxysg:access:file)
Subject Account	Fahad3315 (Recorded Source IP: 10.55.66.777)
Analysis Period	2021-11-16, 09:52:33 – 11:42:38 (Asia/Riyadh, UTC+3)
Investigator	Aljoharah Alqahtani
Prepared For	Cybersecurity Incident Response Team & Investigation Committee
Report Status	Final — Log-Source-Limited Analysis

1. Executive Summary

Proxy telemetry for user Fahad3315 on 2021-11-16 reveals an active security-verification pattern between 10:03 and 11:42 (UTC+3), characterized by repeated queries to VirusTotal and HaveIBeenPwned (HIBP), interleaved with a perimeter-blocked attempt to access a known defacement repository (zone-h.org).

This telemetry establishes an observable timeline of user activity connecting the account to the broader incident window. Because this dataset is restricted strictly to outbound web-proxy logs, it cannot confirm or refute claims of identity impersonation or outbound phishing emails — attribution of unauthorized email activity requires the supplementary mail-gateway and identity audit records detailed in Section 7.

2. Scope and Evidence Source

Total records analyzed	26 log entries (bluecoat:proxysg:access:file)
User & source host	Single authenticated user (Fahad3315) mapped to source IP 10.55.66.777
Proxy collector / auth group	Collector 10.20.30.40 Group: stc\proxylanusers
Distinct destinations	haveibeenpwned.com, www.virustotal.com, www.zone-h.org, youtube.com, www.youtube.com

Background Traffic Excluded: 12 of the 26 records correspond to routine YouTube streaming sessions (09:52–09:56 and 11:18–11:22). This activity was evaluated as non-malicious background browsing, bears no relation to the security-verification pattern, and is excluded from the incident timeline in Section 3 to keep the analysis focused.

Data Integrity Flag: The recorded source IP (10.55.66.777) is an invalid IPv4 address (octet “777” exceeds 255). This must be flagged to log engineering — it suggests log scrubbing, an export artifact, or network-level masking, and reduces confidence in host-level attribution.

Inspection constraints: with the exception of zone-h.org, all sessions were TLS-tunneled (CONNECT/TUNNELED on TCP 443). The proxy captured session metadata, transfer volumes, and timestamps, but payload content, search queries, and submitted hashes/URLs remained encrypted.

3. Incident Timeline of Events

Restricted to security-relevant queries and policy-violation events; routine background browsing is excluded per Section 2.

#	Time (UTC+3)	Destination	Method / Result	Bytes In	Category
1	10:03:34–36	www.virustotal.com	Allowed / CONNECT (×3) & TUNNELED	2,051	Online Services
2	10:07:37	www.virustotal.com	Allowed / TUNNELED	3,152	Online Services
3	10:08:08	www.virustotal.com	Allowed / TUNNELED	644,251	Online Services
4	11:37:32	haveibeenpwned.com	Allowed / CONNECT	39	Technology/Internet
5	11:38:06–08	www.virustotal.com	Allowed / CONNECT (×2)	78	Online Services
6	11:39:24–26	www.zone-h.org (incl. /favicon.ico)	BLOCKED / TCP_DENIED (403)	1,842	Hacking
7	11:42:08	www.virustotal.com	Allowed / TUNNELED	2,169	Online Services
8	11:42:19	www.virustotal.com	Allowed / TUNNELED	26,571	Online Services
9	11:42:38	haveibeenpwned.com	Allowed / TUNNELED	106,910	Technology/Internet

4. Technical Analysis & Findings

Confidence levels below reflect confidence in the behavioral interpretation drawn from the evidence, not the certainty of the underlying log data — all cited figures (byte counts, timestamps, categories) are verified as recorded.

Finding 1 — Sequential Cross-Verification Between HIBP and VirusTotal

Interpretation Confidence: **Medium**

Evidence: Timeline entries #4 through #9 (11:37:32–11:42:38).

Analysis: Within a 5-minute window, the user initiated multiple queries to VirusTotal bracketed by two distinct sessions to HaveIBeenPwned. This behavior is characteristic of an individual checking whether an account or credential was breached while simultaneously inspecting potentially malicious URLs or files. The pattern supports two competing hypotheses: (a) a user actively responding to a perceived security scare or phishing attempt, or (b) an unauthorized operator running situational reconnaissance.

Finding 2 — Blocked Cleartext HTTP Request to a Hacking/Defacement Portal (zone-h.org)

Interpretation Confidence: **Low–Medium**

Evidence: Timeline entry #6 (11:39:24–26): two consecutive denied requests (403 Forbidden), successfully blocked under the Hacking policy category.

Analysis: These are the only cleartext HTTP (port 80) requests in the entire dataset — every other session used HTTPS (port 443) — which is why the full request path and referer header are visible for this destination alone. The initial request carried `cs_Referer = https://www.google.com/`, indicating navigation via a search-engine result rather than a direct link embedded in an email or document.

Finding 3 — Anomalous VirusTotal Response Volume at 10:08:08

Interpretation Confidence: Low

Evidence: Timeline entry #3: bytes_in = 644,251, versus a 1.9–26.5 KB range for other VirusTotal sessions in the dataset.

Analysis: This payload volume is technically consistent with downloading an extensive multi-engine scan report or detailed threat-intelligence summary rather than an empty landing page. Because the session was TLS-encrypted, the specific hash, domain, or IP evaluated cannot be extracted from this log source alone.

Finding 4 — Rapid Service Transitions with a Consistent Interactive Browser Fingerprint**Interpretation Confidence: Low**

Evidence: Timeline entries #8–9: a 19-second gap between VirusTotal (11:42:19) and HIBP (11:42:38). All 12 records in the dataset carrying a User-Agent header show an identical signature — standard Chrome 95 on Windows 10 x64 — with no variation anywhere.

Analysis: The short transition interval points to rapid, manual tab-switching by an interactive user reacting to unfolding events. The uniform browser signature across all sessions, with no automated-scripting indicators (e.g., Python-requests, cURL, headless browsers), corroborates manual operation from a standard workstation rather than a scripted process, though a User-Agent string alone can be spoofed and is not conclusive proof.

5. Incident Hypothesis & Claim Verification

Reported Claim	Investigative Assessment	Telemetry Basis & Rationale
Fahad visited HIBP twice, with VirusTotal activity in between	Confirmed	Directly recorded in telemetry — Timeline entries #4–9 (11:37:32–11:42:38)
Account sent phishing emails impersonating Fahad	Unaddressed by Log Source	Dataset is limited to web-proxy traffic. Validating outbound email transmission requires Exchange/M365 message tracing and SMTP headers
Other employees interacted with malicious messages	Unaddressed by Log Source	Dataset contains telemetry for a single host/account only; interaction by other organizational users cannot be measured here
Host or account compromise	Indeterminate	Web telemetry captures security-verification activity and one blocked request, which are circumstantial and equally indicative of legitimate user self-investigation

6. Data Limitations

- Scope restriction: only 26 records covering ~1 hour 50 minutes for a single user; no baseline period exists for behavioral anomaly comparison.
- Payload blindness: TLS-tunneled traffic (port 443) prevents visibility into queried hashes, URLs, search terms, and VirusTotal detection scores.
- Missing telemetry disciplines: complete absence of EDR telemetry, Active Directory / Entra ID sign-in logs, and mail-gateway tracking.
- Source IP malformation: the invalid IP format (10.55.66.777) prevents definitive physical network-jack or DHCP-lease binding.

7. Actionable Recommendations

- Mail Gateway & Message Trace: pull Exchange / M365 Message Trace logs for Fahad3315 across 2021-11-16 to identify actual outbound message timestamps, sender IP, recipient lists, and SPF/DKIM validation results.
- Identity & Authentication Audit: review Azure AD / Entra ID sign-in logs and Unified Audit Logs (UAL) for Fahad3315 to detect unauthorized concurrent logins, impossible-travel alerts, or new inbox-forwarding rules.
- Attachment / IOC Correlation: extract file hashes or URLs identified from the reported phishing emails and correlate them against the VirusTotal session timing identified in this report (10:03–10:08 and 11:38–11:42).
- Log Pipeline Remediation: escalate the 10.55.66.777 octet anomaly to SIEM/log engineering to identify root cause (e.g., regex error, log masking, or synthetic staging).

In summary, this telemetry substantiates that user Fahad3315 was actively conducting security-checking procedures within a tight timeframe, alongside an isolated, blocked attempt to reach an external defacement database via a search-engine result. These artifacts establish investigative activity on the endpoint but do not, on their own, constitute forensic proof of malicious intent or account takeover; formal incident attribution should remain bounded until the cross-telemetry correlations above are completed.